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# An important role for triglyceride in regulating spermatogenesis

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## Abstract

*Drosophila* is a powerful model to study how lipids affect spermatogenesis. Yet, the contribution of neutral lipids, a major lipid group which resides in organelles called lipid droplets (LD), to normal sperm development is largely unknown. Emerging evidence suggests that LD are present in the testis and that loss of neutral lipid-and LD-associated genes causes subfertility; however, key regulators of testis neutral lipids and LD remain unclear. Here, we show that LD are present in early-stage somatic and germline cells within the *Drosophila* testis. We identified a role for triglyceride lipase *brummer* (*bmm*) in regulating testis LD, and found that whole-body loss of *bmm* leads to defects in sperm development. Importantly, these represent cell-autonomous roles for *bmm* in regulating testis LD and spermatogenesis. Because lipidomic analysis of *bmm* mutants revealed excess triglyceride accumulation, and spermatogenic defects in *bmm* mutants were rescued by genetically blocking triglyceride synthesis, our data suggest that *bmm*-mediated regulation of triglyceride influences sperm development. This identifies triglyceride as an important neutral lipid that contributes to *Drosophila* sperm development, and reveals a key role for *bmm* in regulating testis triglyceride levels during spermatogenesis.


### eLife assessment

This **important** study shows that lipid degradation is critical for spermatogenesis, with data supporting relevance of this finding across phyla. The authors contribute to a growing realization that lipid droplets have critical roles during differentiation and can influence cell fate, and use **convincing** methods to analyze the effects of loss of the lipase Brummer on germ line differentiation. This paper will be of interest to developmental and cell biologists working on gametogenesis.

## Introduction

Lipids play an essential role in regulating spermatogenesis across animals [(1)–(4)]. Studies in *Drosophila* have illuminated key roles for multiple lipid species in regulating sperm development [(5)–(7)]. For example, phosphatidylinositol and its phosphorylated derivatives participate in diverse aspects of *Drosophila* spermatogenesis including meiotic cytokinesis (1; 11), somatic cell differentiation (12), germline and somatic cell polarity maintenance [(13)–(16)], and germline stem cell (GSC) maintenance and proliferation (17). Membrane lipids also influence sperm development (18; 19), whereas fatty acids play a role in processes such as meiotic cytokinesis (20) and sperm individualization (21; 22). While these studies suggest key roles for membrane lipids and fatty acids during *Drosophila* spermatogenesis, some of which are conserved in mammals [(23)–(25)], much less is known about how neutral lipids contribute to spermatogenesis.

Neutral lipids are a major lipid group that includes triglyceride and cholesterol ester, and reside within specialized organelles called lipid droplets (LD) (26). LD are found in diverse cell types (*e.g*. adipocytes, muscle, liver, glia, neurons) (27; 28; 26), and play key roles in maintaining cellular lipid homeostasis. In nongonadal cell types, correct regulation of LD contributes to cellular energy production [(29)–(31)], sequestration and redistribution of lipid precursors [(32)–(36)], and regulation of lipid toxicity [(37)–(39)]. The importance of LD to normal cellular function in nongonadal cell types is shown by the fact that dysregulation of LD causes defects in cell differentiation, survival, and energy production (26; 37; 40; 41). In the testis, much less is known about the regulation and function of neutral lipids and LD, and how this regulation affects sperm development.

Multiple lines of evidence suggest a potential role for neutral lipids and LD during spermatogenesis. First, genes that encode proteins associated with neutral lipid metabolism and LD are expressed in the testis across multiple species [(42)–(44)]. Second, testis LD have been identified in mammals and flies under both normal physiological conditions (27; 48) and after mitochondrial stress (49). Third, loss of genes associated with neutral lipid metabolism and LD cause subfertility phenotypes in both flies and mammals (27; 52). While studies suggest that mammalian testis LD contribute to steroidogenesis (53; 54), the spatial, temporal, and cell-type specific requirements for neutral lipids and LD in the testis have not been explored in detail in any animal. It remains similarly unclear which genes are responsible for regulating neutral lipids and LD during spermatogenesis.

To address these knowledge gaps, we used *Drosophila* to investigate the regulation and function of neutral lipids and LD during sperm development. Our detailed analysis of spermatogenesis under normal physiological conditions revealed the presence of LD in early-stage somatic and germline cells in the testis. We identified triglyceride lipase *brummer* (*bmm*) as a regulator of testis LD, and showed that this represents a cell-autonomous role for *bmm*. Importantly, we found that the *bmm*-mediated regulation of testis LD was significant for spermatogenesis, as both whole-body and cell-autonomous loss of *bmm* caused defects in sperm development. Given that our lipidomic analysis revealed an excess accumulation of triglyceride in animals lacking *bmm*, and that genetically blocking triglyceride synthesis rescued many spermatogenic defects associated with *bmm* loss, our data suggests that *bmm*-mediated regulation of triglyceride is important for normal *Drosophila* sperm development. This reveals previously unrecognized roles for neutral lipids such as triglyceride in regulating spermatogenesis, and for *bmm* in regulating sperm development under normal physiological conditions. Together, these findings advance knowledge of the regulation and function of neutral lipids during spermatogenesis.

# Results

## Lipid droplets are present in early-stage somatic and germline cells

We previously reported the presence of small circular punctae (<1 μm) corresponding to LD near the apical tip of the testis (27). We confirm these results in $w^{1118}$ males using neutral lipid stain BODIPY (4,4-Difluoro-1,3,5,7,8-Pentamethyl-4-Bora-3a,4a-Diaza-*s*-Indacene) (Figure 1A). Importantly, we reproduced this spatial distribution of LD in two independent genetic backgrounds and at two additional ages (Figure 1B,1C). In all cases, LD were in a testis region that contains stem cells and early-stage somatic and germline cells (Figure 1A-A', arrows), and in the hub, an organizing center and stem cell niche in the *Drosophila* testis (Figure 1A''-A''', arrows) (55). LD were largely absent from the testis region occupied by spermatocytes (Figure 1A and A', arrowheads). While LD may contain multiple neutral lipid species(56), cholesterol-binding fluorescent polyene antibiotic filipin III did not detect cholesterol within testis LD (Figure S1A), suggesting triglyceride is the main neutral lipid in *Drosophila* testis LD.

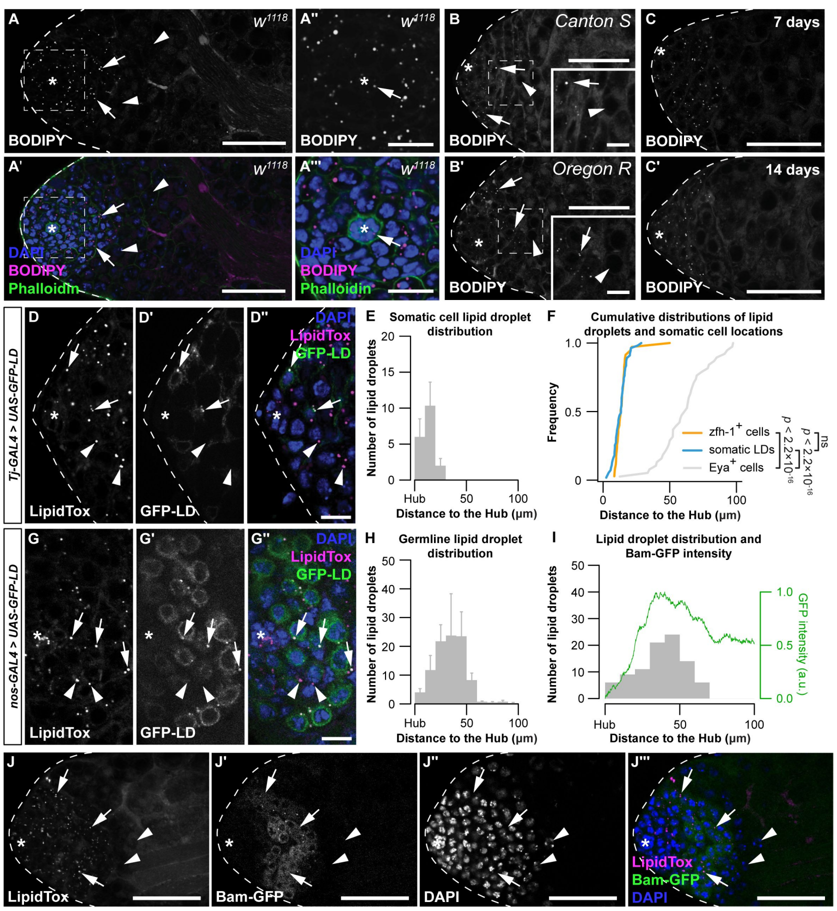



**Figure 1**

### Lipid droplets are present in early-stage somatic and germ cells.

(A) Testis lipid droplets (LD) in $w^{1118}$ animals visualized with neutral lipid dye BODIPY. (A,A') Scale bar=50 μm; (A'',A''') scale bar=15 μm. Asterisk indicates hub in all images. Arrows point to LD; arrowheads point to spermatocytes in A,B. (B) Testis LD visualized with BODIPY in newly-eclosed males from two wild-type genotypes. Scale bars: main image=50 μm; inset image=10 μm. (C) Testis LD from $w^{1118}$ animals at different times post-eclosion. Scale bars=50 μm. (D) Testis LD visualized with LipidTox Red in animals with somatic cell overexpression of GFP-LD (*Tj-GAL4>UAS-GFP-LD*). GFP-and LipidTox Red-positive punctae are somatic LD (D–D'' arrows); LipidTox punctae without GFP indicate germline LD (D–D'' arrowheads). Scale bars=10 μm. (E) Histogram showing the spatial distribution of somatic cell LD; error bars represent standard error of the mean (SEM). (F) Cumulative frequency distributions of somatic LD (blue line, data reproduced from E), zfh-1-positive somatic cells (zfh-1$^{+}$ cells, orange line), and Eya-positive somatic cells (Eya$^{+}$ cells, grey line). (G) Testis LD visualized with LipidTox Red in males with germline overexpression of GFP-LD (*nos-GAL4>UAS-GFP-LD*). GFP-and LipidTox Red-positive punctae indicate germline LD (arrows); LipidTox punctae without GFP indicate non-germline LD (arrowheads). Scale bars=10 μm. (H) Histogram representing the spatial distribution of LD within the germline; error bars represent SEM. (I) Histogram representing the spatial distribution of LD and GFP fluorescence (green line) (arbitrary units, a.u.) in a representative testis of a *bam-GFP* animal (panel J). (J) Testis LD in a *bam-GFP* animal; arrows point to LD and arrowheads point to spermatocytes. Scale bar=50 μm. See also Supplemental Figure 1.

*Drosophila* spermatogenesis requires the codevelopment and differentiation of two cell lineages, the germline and the somatic cells (57). To identify LD in each lineage, we used the GAL4/UAS system to overexpress a transgene in which GFP is fused to the LD-targeting motif of motor protein Klarsicht (58) (*UAS-GFP-LD*). We targeted *UAS-GFP-LD* to somatic cells with *Traffic jam* (*Tj*)*-GAL4* and to the germline using *nanos (nos)-GAL4*; LD were visualized using neutral lipid dye LipidTox. We found LD in the somatic cells of 0-day-old males (Figure 1D), and showed that the majority of somatic LD were <30 μm from the hub (Figure 1E). Because the somatic LD distribution coincided with a marker for somatic stem cells and their immediate daughter cells (Zinc finger homeodomain 1, Zfh-1) (Figure 1F; two-sample Kolmogorov-Smirnov test) (59), but not with a marker for late somatic cells (Eyes absent, Eya) (12; 60), our data suggests LD are present in early somatic cells. In the germline, GFP punctae corresponding to LD were found near the apical tip of the testis in 0-day-old males (Figure 1G,H). We found that the disappearance of germline LD coincided with peak expression of a GFP reporter that reflects the expression of Bag-of-marbles (Bam) protein in the testis (Bam-GFP) (61) (Figure 1I,1J). Because peak Bam expression signals the last round of transient amplifying mitotic cell cycle prior to the germline's transition into the meiotic cell cycle [(62)–(64)], our data suggests that germline LD, like somatic LD, are present at early stages of germline development.

## *brummer* plays a cell-autonomous role in regulating testis lipid droplets

*Adipose triglyceride lipase* (*ATGL*) is a critical regulator of neutral lipid metabolism and LD [(65)–(74)]. Loss of *ATGL* in many cell types triggers LD accumulation, and *ATGL* overexpression decreases LD number (30; 67; 68; 71; 73; 73; 75; 76). Given that the *Drosophila ATGL* homolog *brummer* (*bmm*) regulates testis LD induced by mitochondrial stress (49), we explored whether *bmm* regulates testis LD under normal physiological conditions. We first examined *bmm* expression in the testis by isolating this organ from flies in which a *bmm* promoter fragment drives *GFP* expression (*bmm-GFP*). Indeed, *bmm-GFP* accurately reproduces changes to *bmm* mRNA levels (77). GFP expression was present in the germline of *bmm-GFP* testes, and we found germline GFP levels were higher in spermatocytes than at earlier stages of sperm development (Figure 2A,2B; one-way ANOVA with Tukey multiple comparison test). Supporting this, our analysis of a publicly available single-cell RNA sequencing data set of the male reproductive organ (78) suggested a similar trend in *bmm* mRNA levels between different stages of germline (Figure S2A,S2B) and somatic cell (Figure S2C,S2D) development. Importantly, germline GFP levels were negatively correlated with testis LD in *bmm-GFP* flies (Figure 2A,2C), suggesting regions with higher *bmm* expression had fewer LD.

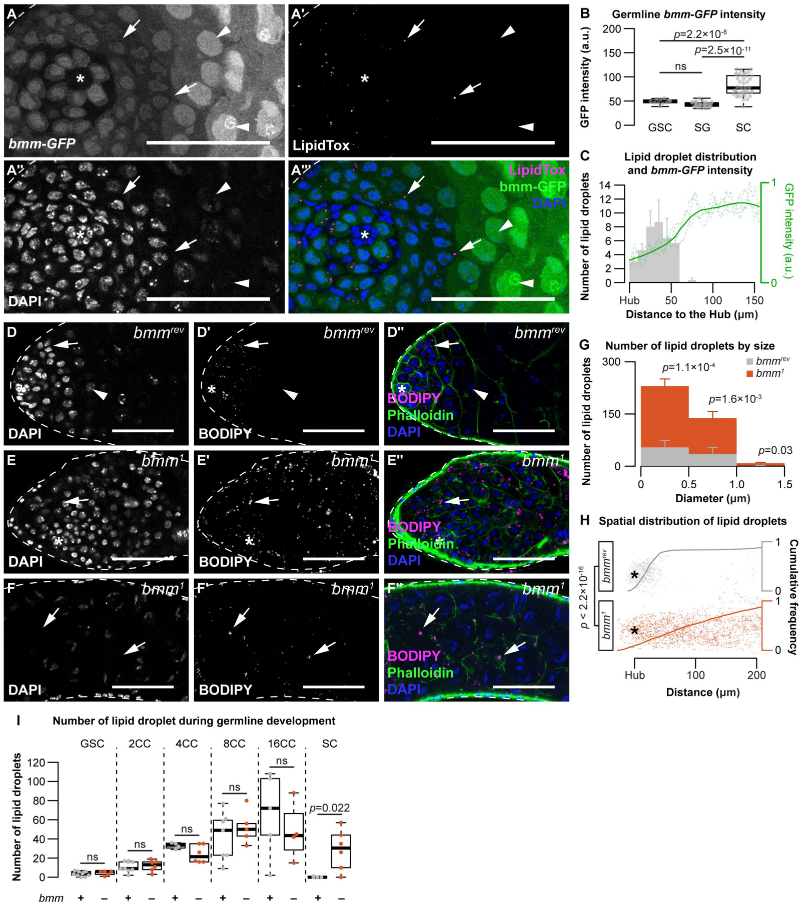


**Figure 2**

### *bmm* regulates testis lipid droplets in a cell-autonomous manner.

(A) Testis lipid droplets (LD) indicated by LipidTox Red in *bmm-GFP* animals. Arrows point to LD in all images. Arrowheads point to spermatocytes. Scale bars=50 μm. Asterisks indicate the hub in all images. (B) Quantification of nuclear GFP intensity in testes isolated from *bmm-GFP* animals (n=3). Germline stem cell (GSC), spermatogonia (SG), spermatocyte (SC). (C) Spatial distribution of LD (grey histogram) and GFP expression (green line) in testes from *bmm-GFP* animals as a function of distance from the hub (n=3). (D,E) LD near the apical region of the testis in *bmm^rev^* (D) or *bmm^1^* (E) animals. (F) LD further away from the apical tip in *bmm^1^* animals. (D–F) Scale bars=50 μm. (G) Histogram representing testis LD size distribution in *bmm^rev^* (grey) and *bmm^1^* (orange). (H) Apical tip of the testes is at the left of the graph; individual dots represent a single LD and its relative position to the hub marked by an asterisk. Cumulative frequency distribution of the distance between LDs and the apical tip of the testes are drawn as solid lines. (I) Number of testis LD in *bmm^rev^* (grey) or *bmm^1^* (orange) in FLP-FRT clones 3 days post-clone induction; dots represent measurements from a single clone. The number of cells in each cyst (CC) counted is indicated. See also Supplemental Figure 2.

To test whether *bmm* regulates testis LD, we compared LD in testes from 0-day-old males carrying a loss-of-function mutation in *bmm* (*bmm$^{1}$*) to control male testes (*bmm$^{rev}$*)(67). *bmm$^{1}$* testes had significantly more LD across all LD sizes compared with control males (Figure 2D–2G; Welch two-sample t-test with Bonferroni correction), and showed a significantly expanded LD distribution (Figure 2D–2F,2H; two-sample Kolmogorov-Smirnov test). This suggests *bmm* normally restricts LD to the region near the apical tip of the testis, a role we confirm in both somatic and germline lineages (Figure S2E–S2H). Importantly, after inducing homozygous *bmm$^{1}$* or *bmm$^{rev}$* clones in the testes using *FLP-FRT* system(79), we found *bmm$^{1}$* spermatocyte clones had significantly more LD at 3 days post-clone induction (Figure 2I; Welch two-sample t-test), a stage at which LD were absent from *bmm$^{rev}$* clones. This indicates a previously unrecognized cell-autonomous role for *bmm* in regulating testis LD, a role we were unable to assess in somatic cells as we recovered no *bmm$^{1}$* somatic cell clones.

## brummer plays a cell-autonomous role in regulating germline development

To determine the physiological significance of *bmm*-mediated regulation of testis LD, we investigated testis and sperm development in males without *bmm* function. In 0-day-old *bmm*$^{1}$ males reared at 25°C, testis size was significantly smaller than in age-matched *bmm*$^{rev}$ controls (Figure S3A; Welch two-sample t-test), and the number of spermatid bundles was significantly lower (Figure S3B; Kruskal-Wallis rank sum test). Defects in testis size and sperm development were also observed in 14-day-old *bmm*$^{1}$ males (Figure S3C,S3D Welch two-sample t-test). When the animals were reared at 29°C, a temperature that exacerbates spermatogenesis defects associated with changes in lipid metabolism (21), *bmm*$^{1}$ phenotypes were more pronounced (Figure 3A-3C). This suggests loss of *bmm* affects testis development and spermatogenesis. Because similar phenotypes are observed in male mice without ATGL (52), and supplementing the diet of *bmm*$^{1}$ males with medium-chain triglycerides (MCT) partially rescues the testis and spermatogenic defects we observed in flies (Figure S3E,S3F; one-way ANOVA with Tukey multiple comparison test), as it does in mice (52; 80), our data suggests flies are a good model to study how *bmm*/*ATGL* influences sperm development.

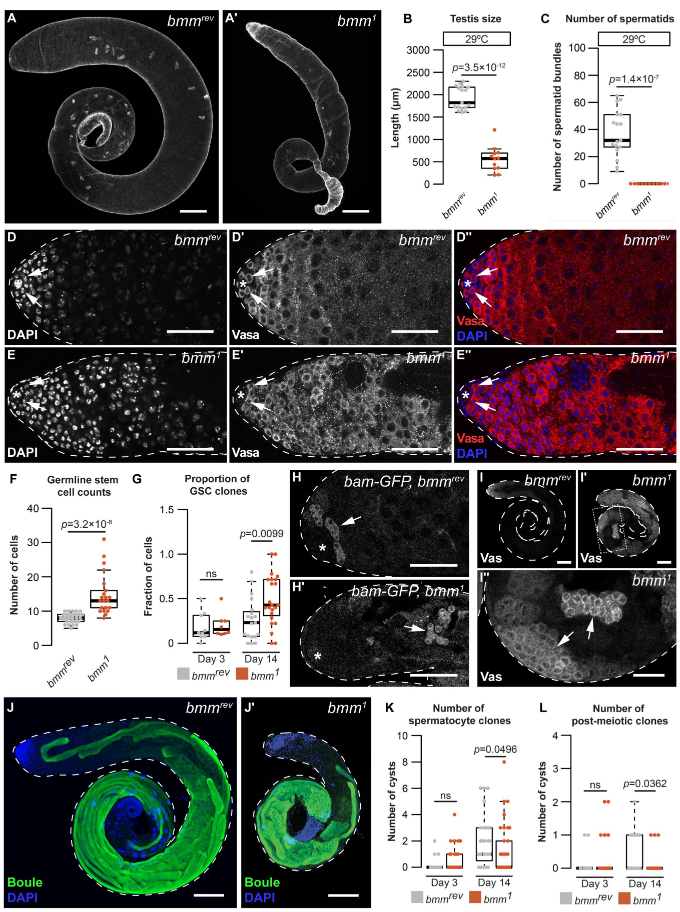



**Figure 3**

### A cell-autonomous role for *bmm* in regulating spermatogenesis.

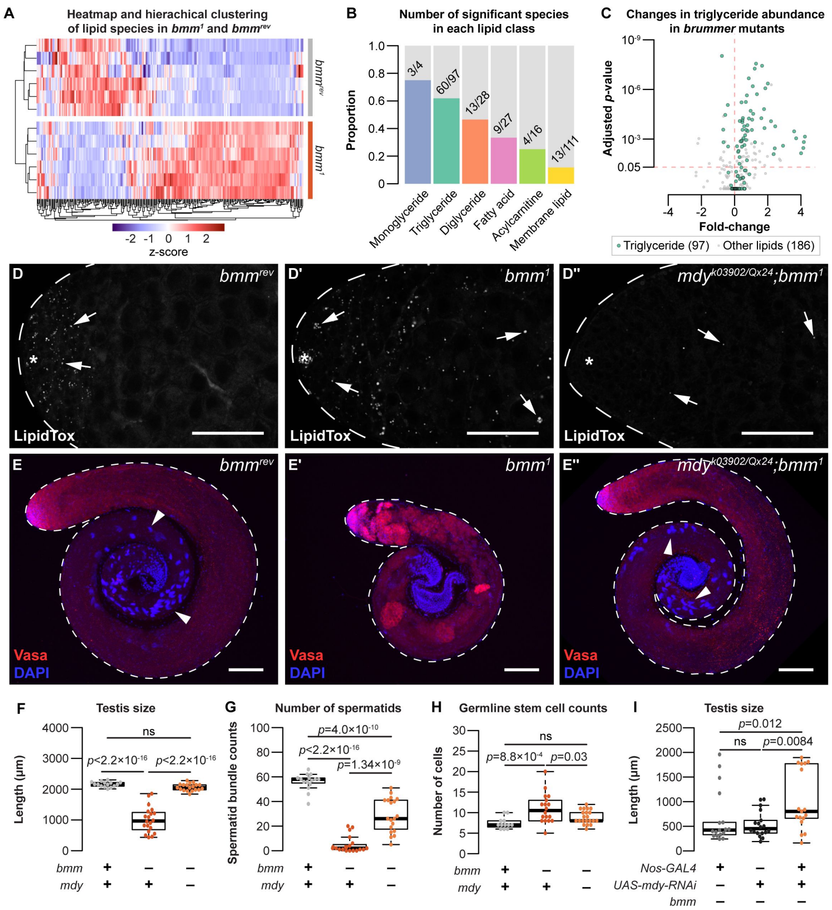
Testes isolated from *bmm*$^{rev}$ (A) and *bmm*$^{1}$ (A') animals raised at 29°C stained with phalloidin. Scale bars=100 µm. (B) Testis size in *bmm*$^{1}$ and *bmm*$^{rev}$ animals raised at 29°C. (C) Spermatid bundle number in *bmm*$^{1}$ and *bmm*$^{rev}$ testes from animals reared at 29°C. (D,E) Representative images of *bmm*$^{rev}$ (D) or *bmm*$^{1}$ (E) testes stained with DAPI and anti-Vasa antibody. Arrows indicate germline stem cells (GSC). Scale bar=50 µm. The hub is marked by an asterisk in all images. (F) GSC number in *bmm*$^{1}$ and *bmm*$^{rev}$ testes. (G) Proportion of GSCs that were either *bmm*$^{1}$ or *bmm*$^{rev}$ clones at 3 and 14 days post-clone induction. (H) Representative images of *bmm*$^{rev}$ (H) and *bmm*$^{1}$ (H') testes carrying *bam-GFP*; data quantified in Figure S3J. Arrows indicate regions with high Bam-GFP. Scale bars=50 µm. (I) Representative images of *bmm*$^{rev}$ (I) or *bmm*$^{1}$ (I',I'') testes stained with anti-Vasa antibody. Arrows indicate Vasa-positive cysts in *bmm*$^{1}$ testis. Panel I'' is magnified from the boxed region in I'. (I,I') Scale bars=100 µm; (I'') scale bar=50 µm. (J) Maximum projection of *bmm*$^{rev}$ (J) or *bmm*$^{1}$ (J') testes stained with anti-Boule antibody (green) and DAPI (blue). Scale bars=100 µm. Number of *bmm*$^{1}$ and *bmm*$^{rev}$ spermatocyte clones (K) or post-meiotic clones (L) at 3 and 14 days post-clone induction. See also Supplemental Figure 3.

To explore spermatogenesis in *bmm*$^{1}$ animals, we used germline-specific marker Vasa to visualize the germline in the testes of *bmm*$^{1}$ and *bmm*$^{rev}$ males (Figure 3D,3E) (81). We observed a significant increase in the number of germline stem cells (GSC) (Figure 3F; Kruskal-Wallis rank sum test) and higher variability in GSC number in *bmm*$^1$ males ($p$=5.7×10$^{-12}$ by F-test). Given that GSC number is affected by hub size and GSC proliferation (82; 83), we monitored both parameters in *bmm*$^1$ and *bmm*$^{rev}$ controls. While hub size in *bmm*$^1$ testes was significantly larger than in testes from *bmm*$^{rev}$ controls (Figure S3G,S3H; Welch two-sample t-test), the number of phosphohistone H3-positive GSC, which indicates proliferating GSC, was unchanged in *bmm*$^1$ animals (Figure S3I; Kruskal-Wallis rank sum test). While this indicates a larger hub may partly explain *bmm*'s effect on GSC number, *bmm* also plays a cell-autonomous role in regulating GSC, as we recovered a higher proportion of *bmm*$^1$ clones in the GSC pool compared with *bmm*$^{rev}$ clones at 14 days after clone induction (Figure 3G; Welch two-sample t-test).

Beyond GSC, we uncovered additional spermatogenesis defects in *bmm*$^1$ testes. Peak Bam-GFP expression in testes from 0-day-old *bmm*$^1$ and *bmm*$^{rev}$ males showed that GFP-positive cysts with were significantly further away from the hub in *bmm*$^1$ testes (Figure 3H,S3J; Welch two-sample t-test). Indeed, 15/18 *bmm*$^1$ testes contained Vasa-positive cysts with large nuclei in the distal half of the testis (Figure 3I, arrowheads), a phenotype not present in *bmm*$^{rev}$ testes (0/8) ($p$=0.0005 by Pearson's Chi-square test). Because these phenotypes are also seen in testes with differentiation defects (13; 84), we recorded the stage of sperm development reached by the germline in *bmm*$^1$ testes. Most *bmm*$^1$ testes contained post-meiotic cells in males raised at 25°C (Figure S3K); however, germline development did not progress past the spermatocyte stage in most *bmm*$^1$ testes from animals raised at 29°C (Figure S3K). Testes from *bmm*$^1$ males reared at 25°C also had a smaller Boule-positive area (Figure 3J,S3L; Welch two-sample t-test), and fewer individualization complexes and waste bags (Figure S3M,S3N; Kruskal-Wallis rank sum test). Together, these data indicate loss of *bmm* delays germline development. Because we recovered fewer *bmm*$^1$ spermatocyte and spermatid clones 14 days after clone induction (Figure 3K,3L; Kruskal-Wallis rank sum test), this effect on germline development represents a cell-autonomous role for *bmm*.

## brummer-dependent regulation of testis triglyceride levels affects spermatogenesis

*ATGL* catalyzes the first and rate-limiting step of triglyceride hydrolysis (73; 85; 86). Loss of this enzyme or its homologs leads to excess triglyceride accumulation (27; 30; 67; 73; 75) and shifts in multiple lipid classes (66; 89). To determine how loss of *bmm* affects spermatogenesis, we carried out mass spectrometry (MS)-based untargeted lipidomic profiling of *bmm*$^1$ and *bmm*$^{rev}$ males. Hierarchical clustering of lipid species suggests that *bmm*$^1$ and *bmm*$^{rev}$ males show distinct lipidomic profiles (Figure 4A). Overall, we detected 2464 and 1144 lipid features with high quantitative confidence in positive and negative ion modes, respectively. By matching experimental *m/z*, isotopic ratio, and tandem MS spectra to lipid libraries, we confirmed 293 unique lipid species (Supplemental table 1). We found 107 lipids had a significant change in abundance between *bmm*$^1$ and *bmm*$^{rev}$ males (padj<0.05): 85 species were upregulated in *bmm*$^1$ males and 22 lipid species were downregulated. Among differentially regulated species from different lipid classes, triglyceride had the largest residual above expected proportion ($p$=5.00×10$^{-4}$ by Pearson's Chi-squared test). This suggests triglyceride is the lipid class most affected by loss of *bmm* (Figure 4B,4C).




## Figure 4

# Loss of *bmm* disrupts triglyceride homeostasis and leads to spermatogenic defects.

(A) Hierarchical clustering of lipid species detected in $bmm^{rev}$ and $bmm^{1}$ animals. (B) Histograms showing the proportion of species in each lipid class with different levels between $bmm^{1}$ and $bmm^{rev}$. Numbers on histograms indicate the number of species with differences in abundance. (C) Volcano plot showing fold change in abundance of triglycerides (green; 97 species) and non-triglyceride lipids (grey; 186 species) in our dataset. (D) Arrows indicate testis LD stained with LipidTox Red in $bmm^{rev}$ (D), $bmm^{1}$ (D′), or $mdy^{QX25/k03902}$;$bmm^{1}$ (D′′) animals. (E) Whole testes isolated from $bmm^{rev}$ (E), $bmm^{1}$ (E′), or $mdy^{QX25/k03902}$;$bmm^{1}$ (E′′) animals stained with anti-Vasa antibody (red) and DAPI (blue). Arrowheads indicate spermatid bundles. Scale bars=100 µm. (F) Testis size in $bmm^{rev}$, $bmm^{1}$, and $mdy^{QX25/k03902}$;$bmm^{1}$ animals. Spermatid bundles (G) and number of germline stem cells (H) in $bmm^{rev}$, $bmm^{1}$, and $mdy^{QX25/k03902}$;$bmm^{1}$ animals. (I) Testis size in animals with germline-specific *mdy* knockdown (*nos-GAL4>mdy RNAi*; $bmm^{1}$) compared with controls (*nos-GAL4>+*; $bmm^{1}$ and *+>mdy RNAi*; $bmm^{1}$). See also Supplemental Figure 4.

In $bmm^{1}$ males, most triglyceride species (55/97) were significantly higher. Because we observed a positive correlation between the fold increase in triglyceride abundance with both the number of double bonds ($p=7.52\times10^{-8}$ by Kendall's rank correlation test; Figure S4A) and the number of carbons ($p=2.77\times10^{-10}$ by Kendall's rank correlation test; Figure S4B), our data align well with *bmm/ATGL's* known role in regulating triglyceride levels(67; 68; 73) and its substrate preference of long-chain polyunsaturated fatty acids(85). While we also detected changes in species such as fatty acids, acylcarnitine, and membrane lipids (Figure S4C–S4H), in line with recent *Drosophila* lipidomic data(90; 91), the striking accumulation of triglyceride in $bmm^{1}$ males suggested that excess testis triglyceride in $bmm^{1}$ males may contribute to their spermatogenic defects. To test this, we examined spermatogenesis in $bmm^{1}$ males carrying loss-of-function mutations in *midway* (*mdy*). *mdy* is the *Drosophila* homolog of *diacylglycerol O-acyltransferase 1* (*DGAT1*), and whole-body loss of *mdy* reduces whole-body triglyceride levels[(92)–(94)]. Importantly, testes isolated from males lacking both *bmm* and *mdy* (genotype $mdy^{QX25/k03902}$;$bmm^{1}$) had fewer LD than testes dissected from $bmm^{1}$ males (Figures 4D,S4I; one-way ANOVA with Tukey multiple comparison test).

We found that testes isolated from $mdy^{QX25/k03902}$;$bmm^{1}$ males were significantly larger and had more spermatid bundles than testes from $bmm^{1}$ males (Figure 4E–G; one-way ANOVA with Tukey multiple comparison test). The elevated number of GSC in $bmm^{1}$ male testes was similarly rescued in *mdy*$^{QX25/k03902}$;*bmm*$^{1}$ males (Figure 4H; one-way ANOVA with Tukey multiple comparison test). These data suggest that defective spermatogenesis in *bmm*$^{1}$ males can be partly attributed to excess triglyceride accumulation. Notably, at least some of these defects are cell-autonomous: RNAi-mediated knockdown of *mdy* in the germline of *bmm*$^{1}$ males partially rescued the defects in testis size (Figure 4I; Kruskal-Wallis rank sum test with Dunn's multiple comparison test) and GSC variance (Figure S4J; $p$=4.5 x $10^{-5}$ and 8.2 x $10^{-3}$ by F-test from the GAL4-and UAS-only crosses, respectively). *bmm*-mediated regulation of testis triglyceride therefore plays a previously unrecognized role in regulating sperm development.

# Discussion

In this study, we used *Drosophila* to gain insight into how the neutral lipids, a major lipid class, contribute to sperm development. We describe the distribution of LD under normal physiological conditions in the *Drosophila* testis, and show that LD are present at the early stages of development in both somatic and germline cells. While many factors are known to regulate LD in nongonadal cell types, we reveal a cell-autonomous role for triglyceride lipase *bmm* in regulating testis LD during spermatogenesis. Indeed, our data indicates loss of *bmm* delays germline differentiation leading to an accumulation of early-stage germ cells. These defects in germline differentiation can be partially explained by the excess accumulation of triglyceride in flies lacking *bmm*, as genetically blocking triglyceride synthesis rescues multiple spermatogenic defects in *bmm* mutants. Together, our data reveals previously unrecognized roles for LD and triglycerides during spermatogenesis, and for *bmm* as an important regulator of testis LD and germline development under normal physiological conditions.

One key outcome of our study was increased knowledge of LD regulation and function in the testis. Despite rapidly expanding knowledge of LD in cell types such as adipocytes or skeletal muscle, less is known about how LD influence spermatogenesis under normal physiological conditions. In mammals, testis LD contain cholesterol and play a role in promoting steroidogenesis (95; 96). In flies, we show that LD are present in the testis, and that excess accumulation of these LD affects sperm development. In nongonadal cell types, triglycerides provide a rich source of fatty acids for cellular ATP production, lipid building blocks to support membrane homeostasis and growth, and metabolites that can act as signaling molecules (26). Because ATP production, lipid precursors, and lipid signaling all play roles in supporting normal sperm development (97; 98), future studies will need to determine how each of these processes is affected when excess triglyceride accumulates in testis LD. This will provide critical insight into how triglyceride stored within testis LD contributes to overall cellular lipid metabolism during spermatogenesis. Because of the parallel spermatogenic defects we observed in *bmm* mutants and *ATGL*-deficient mice, we expect that these mechanisms will also operate in other species.

A more comprehensive understanding of neutral lipid metabolism during sperm development will also emerge from studies on the upstream signaling networks that regulate testis LD and triglyceride. Given that we show an important and cell-autonomous role for *bmm* in regulating testis LD and triglyceride, future studies will need to identify factors that regulate *bmm* in the testis. Based on public single-cell RNAseq data and the *bmm-GFP* reporter strain, our data suggest *bmm* mRNA levels are differentially regulated between early and later stages of sperm development. Candidates for mediating this regulation include the insulin/insulin-like growth factor signaling pathway (IIS), Target of rapamycin (TOR) pathway, and nuclear factor κB/Relish pathway (NFκB), as all of these pathways influence *bmm* mRNA levels in nongonadal cell types [(99)–(105)]. Beyond mRNA levels, Bmm protein levels and post-translational modifications may also be differentially regulating during spermatogenesis. For example, studies show that the proteins encoded by *bmm* homologs in other animals are regulated by phosphorylation (106), mediated by kinases such as adenosine monophosphate-activated protein kinase (AMPK) and protein kinase A (PKA) [(107)–(109)]. Importantly, many of these pathways, including IIS, TOR, AMPK, NFκB and possibly PKA influence *Drosophila* sperm development [(110)–(115)]. Identifying the signaling networks that influence *bmm* regulation during sperm development will therefore lead to a deeper understanding of how testis LD and triglyceride are coordinated with physiological factors to promote normal spermatogenesis. Because pathways such as IIS and AMPK, and others, regulate sperm development in other species [(116)–(118)], these insights may reveal conserved mechanisms that govern the regulation of cellular neutral lipid metabolism during sperm development.

## Acknowledgements

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## Author contributions

Conceptualization, C.C. and E.J.R.; Methodology, C.C. and Y.Y.P.; Software, C.C.; Investigation, C.C., H.Y., and Y.Y.P.; Lipidomics, M.A., H.Y., C.W., T.H.; Writing – Original Draft, C.C. and E.J.R.; Writing – Review and Editing, C.C., E.J.R., and Y.Y.P.; Supervision, E.J.R., G.T., and T.H.; Project administration, E.J.R.; Funding Acquisition, E.J.R., G.T., and T.H.

## Declaration of interests

The authors declare no competing interests.

# Materials and methods

## Materials and Resource availability.

*Drosophila* strains and their source are listed in a Key Resources table. Further information and requests for resources and reagents should be directed to, and will be fulfilled by, lead contact Dr. Elizabeth J. Rideout (elizabeth.rideout@ubc.ca).

## Data and Code availability.

All raw data and results of statistical tests reported in this paper are located in Supplementary files 1-4. This paper does not report original code. Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

## Fly husbandry.

Fly stocks were maintained at room temperature in 12:12 hour light:dark cycle. Unless otherwise indicated, all flies were raised at 25°C with a density of 50 larvae per 10 mL fly media. Because this project examines sperm development, we used male flies in all experiments. Fly media contained 20.5 g sucrose (SU10, Snow Cap), 70.9 g Dextrose (SUG8, Snow Cap), 48.5 g cornmeal (AO18006, Snow Cap), 30.3 g baker's yeast (NB10, Snow Cap), 4.55 g agar (DR-820-25F, SciMart), 0.5 g calcium chloride dihydrate (CCL302.1, BioShop Canada), 0.5 g magnesium sulfate heptahydrate (MAG511.1, BioShop Canada), 4.9 mL propionic acids (P1386, Sigma-Aldrich), and 488 µL phosphoric acid (P5811, Sigma-Aldrich) per 1L of media. For diets with medium-or long-chain triglyceride, 4 g of coconut oil (medium chain triglyceride) or olive oil (long chain triglyceride) was added per 100 mL of media described above prior to cooling. Males were collected and dissected within 24 hours of eclosion unless otherwise indicated. Fixations were performed at room temperature with 4% paraformaldehyde (CA11021-168, VWR) in PBS for 20 minutes on a rotating platform followed by washing in PBS twice before staining. Fly strains used in our study are listed in a Key Resources table.

## Testis cell stage classification and measurements.

Cells at an early stage of development (stem cells and early-stage somatic and germline cells) were located in the apical region of the testis, and were identified by their small and dense nuclei(120). GSC were defined as Vasa-positive cells in direct contact with the hub; proliferating GSC were identified as Vasa-positive cells in direct contact with the hub that were also phospho-H3 positive. Cells in the testis region occupied by primary spermatocytes were identified by their large cell size and decondensed chromosome staining occupying three nuclear domains(120). Spermatid bundles were identified by their condensed and needle-shaped nuclei, which roughly corresponds to nuclei with protamine-based chromatin(121). Testis size was measured by quantifying the length of a line drawn down the middle of a testis image; starting from the apical tip of the testis and ending where the testis meets the seminal vesicle.

## FLP-FRT clone induction.

Adult males were collected at 3-5 days post-eclosion and heat-shocked three times at 37°C with a 10 min rest period at room temperature between heat shocks. After heat-shock, the flies were incubated at room temperature until dissection.

## *Immunohistochemistry*.

Fixed samples were rinsed three times with blocking solution containing 0.2% bovine serum albumin (A4503, Sigma-Aldrich), 0.3% Triton-X in PBS, then blocked for 1 hr on a rotating platform at room temperature. During the incubation, the blocking solution was changed every 15 minutes. After blocking, the sample were resuspended in blocking solution with the appropriate concentration of primary antibody (see Key Resources table), and incubated overnight at 4°C. Samples were rinsed three times with blocking solution after removing primary antibody, and blocked for one hour on a rotating platform in blocking solution. Secondary antibody was applied in blocking solution and left on the rotating platform at room temperature for 40 min. The sample was rinsed with blocking solution three more times, and washed four times for 15 min per wash in blocking solution. Testis samples were resuspended in Vectashield mounting media with DAPI (H-1200-10, Vector Laboratory) or SlowFade Diamond mounting media (S36972, Thermo Fisher Scientific) prior to mounting.

## *Lipid droplet staining*.

Fixed testes were briefly permeabilized with 0.1% Triton-X in PBS for 5 min prior to applying phalloidin. For BODIPY staining, samples were suspended in PBS containing 10 μg/mL DAPI (2879083-5mg, PeproTech), 1:500 BODIPY 495/503 (Thermo Fisher Scientific D3922), and 1:1000 phalloidin iFluor647 (ab176759, Abcam) or 1:40 phalloidin TexasRed (T7471, Thermo Fisher Scientific). For staining with LipidTox Red, samples were suspended in PBS containing 10 μg/mL DAPI (2879083-5mg, PeproTech), 1:200 LipidTox Red (H34476, Thermo Fisher Scientific), and 1:1000 phalloidin iFluor647 (ab176759, Abcam). For staining free sterols, samples were prepared as for BODIPY staining with 50 μg/mL filipin in place of BODIPY for 30 min. Samples were incubated on a rotating platform for 40 minutes at room temperature. After incubation, samples were washed twice with PBS, then resuspended in SlowFade Diamond mounting media (Thermo Fisher Scientific S36972) prior to mounting.

## *Image acquisition and processing*.

All images were acquired on a Leica SP5 confocal microscope system with 20X or 40X objectives and quantified with Fiji image analysis software(122).

## *Drosophila lipidomics*.

*Drosophila* extracts were prepared following the previously reported protocol(123). Briefly, 10 *Drosophila* males (~10 mg) were weighed, 300 μL of ice-cold methanol/water mixture (9:1, v:v) was added to these males, and the samples were homogenized with glass beads using a bead beater (mini-beadbeater-16, BioSpec, Bartlesville, Ok, USA). Sample weight was used for sample normalization. Fly lysate was kept at-20°C for 4 hours for protein precipitation. Then, 900 μL of methyl tert-butyl ether was added and the solution was shaken for 5 min to extract lipids. To induce phase separation 285 μL of water was added, followed by centrifugation. The upper layer was separated, dried, and reconstituted in isopropanol/acetonitrile (1:1, v:v) for liquid chromatography-mass spectrometry (LC-MS) analysis. The volume of reconstitution solution was proportional to sample weight for normalization. Quality control (QC) samples were prepared by pooling 20 μL aliquot from each sample. The method blank sample was prepared using an identical workflow but without adding *Drosophila*.

*Drosophila* extracts were analyzed on an UHR-QqTOF (Ultra-High Resolution Qq-Time-Of-Flight) mass spectrometry Impact II (Bruker Daltonics, Bremen, Germany) interfaced with an

Agilent 1290 Infinity II LC Systems (Agilent Technologies, Santa Clara, CA, USA). LC separation was performed using a Waters reversed-phase (RP) UPLC Acquity BEH C18 Column (1.7 µm, 1.0 mm ×100 mm, 130 Å) (Milford, MA, USA) maintained at 30°C. For positive ion mode, the mobile phase A was 60% acetonitrile in water and the mobile phase B was 90% isopropanol in acetonitrile, both containing 5 mM ammonium formate (pH = 4.8, adjusted by formic acid). For negative ion mode, the mobile phase A was 60% acetonitrile in water and the mobile phase B was 90% isopropanol in acetonitrile, both containing 5 mM NH4FA (pH = 9.8, adjusted by ammonium hydroxide). The LC gradient for positive and negative ion modes was set as follows: 0 min, 5% B; 8 min, 40% B; 14 min, 70% B; 20 min, 95% B; 23 min, 95% B; 24 min, 5% B; 33 min, 5% B. The flow rate was 0.1 mL/min. The injection volume was optimized to 2 µL in positive mode and 5 µL in negative mode using QC sample. The ESI source conditions were set as follows: dry gas temperature, 220 °C; dry gas flow, 7 L/min; nebulizer gas pressure, 1.6 bar; capillary voltage, 4500 V for positive mode and 3000 V for negative mode. The MS1 analysis was conducted using following parameters: mass range, 70-1000 m/z; spectrum type: centroid, calculated using maximum intensity; absolute intensity threshold: 250. Data-dependent MS/MS analysis parameters: collision energy: 16-30 eV; cycle time, 3 s; spectra rate: 4 Hz when intensity < $10^4$ and 12 Hz when intensity > $10^5$, linearly increased from $10^4$ to $10^5$. External calibration was applied using sodium formate to ensure the m/z accuracy before sample analysis.

The raw LC-MS data were processed using MS-DIAL (ver. 4.38)(124). The detailed MS-DIAL parameters are: MS1 tolerance, 0.01 Da; MS/MS tolerance, 0.05; mass slice width, 0.05 Da; smoothing method, linear weighted moving average; smoothing level, 3 scans; minimum peak width, 5 scans. Lipid features with high quantitative confidence were selected by the following criteria: retention time was within the gradient elution time (< 23 min); average intensity in QC samples is larger than 5-fold of the intensity in method blank sample. Lipid identification was performed by matching experimental precursor m/z, isotopic ratio and MS/MS spectrum against the LipidBlast libraries embedded in MS-DIAL. To improve the quantification accuracy, the measured MS signal intensities were corrected using serial diluted QC samples following the reported workflow(125).

## Quantification and statistical analysis.

All microscopy images were quantified using Fiji software(122). For lipid droplet counts, a single optical slice through the middle of the testis containing the hub was used with the exception of FLP-FRT experiment where all lipid droplets within a GFP-negative cyst were counted (Figure 2I). All statistical analyses were done using R (obtained from https://cran.r-project.org). With exception of data concerning spatial distribution, and lipidomic data, Shapiro-Wilk test (via *shapiro.test* in base R) was used to assess normality of distribution prior to testing for significance. Kruskal-Wallis rank sum test (from the R package *coin*) and Dunn's test (from the R package *dunn.test*) were used in place of Welch two-sample t-test and Tukey's multiple comparison test when the assumption of normality was not met. For testing differences in variance between two populations, F-test (via *var.test* in base R) was used. For testing differences in spatial distribution, two-sample Kolmogorov-Smirnov test (via *ks.test* in base R) was used. All *p*-values are indicated in figures; extremely small *p*-values are listed as $p<2.2 \times 10^{-16}$.

# Resource table

| REAGENT or RESOURCE | SOURCE | RESOURCE # |
|---|---|---|
| **Antibodies** | | |
| Anti-Vasa (1:200) | Gift from Dr. R. Lehman, MIT | |
| Anti-Eya (1:50) | Developmental Studies Hybridoma Bank (DSHB) | eya10H6 |
| Anti-zfh1 (1:1000) | Gift from Dr. J. Skeath, WUSTL | |
| Anti-boule (1:1000) | Gift from Dr. S. Wasserman, UCSD | |
| Anti-phospho-histone H3 (1:1000) | Millipore Sigma | 05-1354 |
| **Experimental models: *Drosophila melanogaster*** | | |
| *$w^{1118}$* | Bloomington *Drosophila* stock center | 3605 |
| *CantonS* | Bloomington *Drosophila* stock center | 64349 |
| *OregonR* | Bloomington *Drosophila* stock center | 25211 |
| *$bmm^{1}$* | Gift from Dr. R. Kühnlein | |
| *$bmm^{rev}$* | Gift from Dr. R. Kühnlein | |
| *mdy[Qx25], cn[1], bw[1]/CyO, l(2)DTS513[1]* | Bloomington *Drosophila* stock center | 5095 |
| *y[1],w[67c23];P{lacW}Cse1[k03802],mdy[k03902]/CyO* | Bloomington *Drosophila* stock center | 10536 |
| *w[1118];P{GD1749}v6367 (UAS-mdy-RNAi)* | Vienna *Drosophila* resource center | 6367 |

| REAGENT or RESOURCE | SOURCE | RESOURCE # |
|---|---|---|
| *nos-GAL4::VP16* | Bloomington *Drosophila* stock center | 7303 |
| *Tj-GAL4* | Gift from Dr. D. Godt, University of Toronto | |
| *c587-GAL4* | Bloomington *Drosophila* stock center | 67747 |
| Bam-GFP | [40] | |
| *bmm-GFP* | Gift from Dr. K. Kamei[57] | |
| *GFP-LD* | Gift from Dr. M. Welte[36] | |
| *P{neoFRT}82B, bmm[1]* | This study | |
| *P{neoFRT}82B, bmm[rev]* | This study | |
| *bam-GFP, bmm[1]* | This study | |
| *bam-GFP, bmm[rev]* | This study | |
| **Software and algorithms** | | |
| Fiji | https://imagej.net/software/fiji/ | |
| R | https://cran.r-project.org | |

## Supplemental table and files

**Supplemental table 1** – Table showing identified lipid species from untargeted lipidomic analysis.

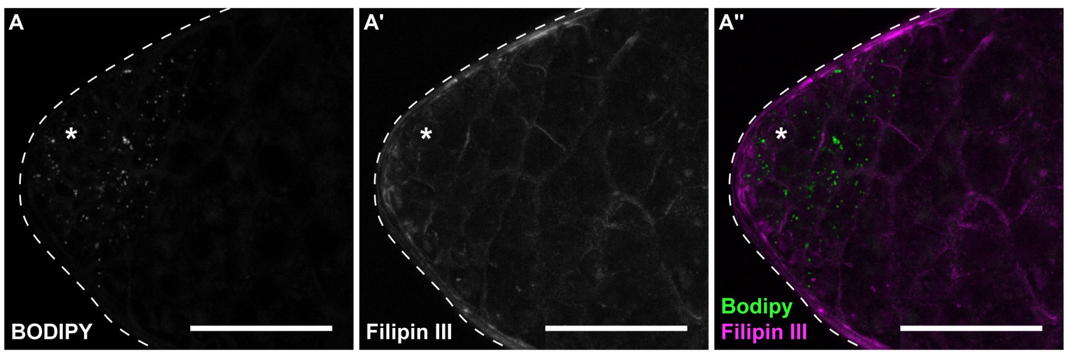
**Supplementary file 1** – Raw data and statistical outputs from Figure 1.

**Supplementary file 2** – Raw data and statistical outputs from Figure 2 and Supplemental figure 2.

**Supplementary file 3** – Raw data and statistical outputs from Figure 3 and Supplemental figure 3.

**Supplementary file 4** – Raw data and statistical outputs from Figure 4 and Supplemental figure 4.

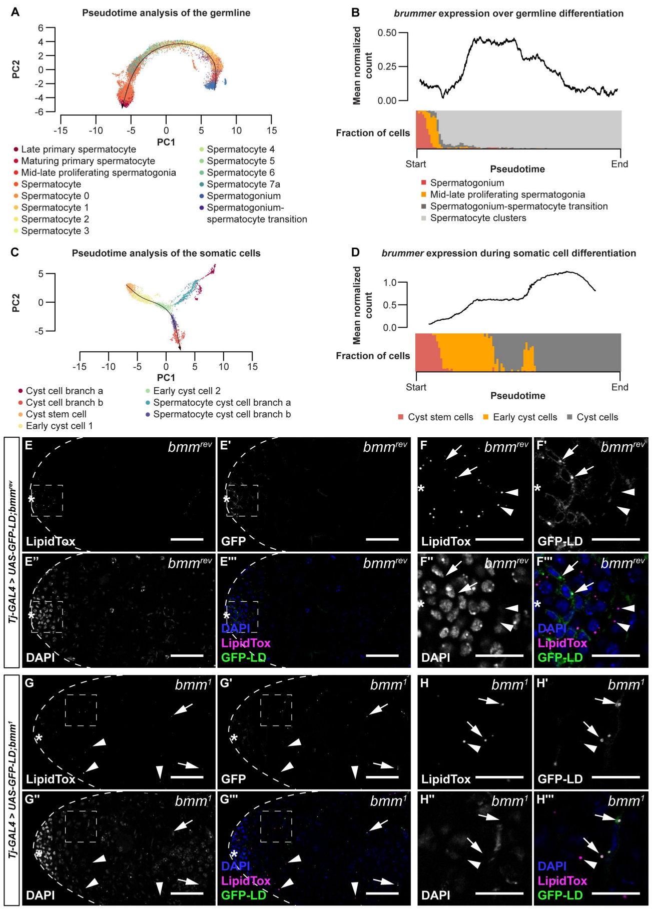
## Supplemental figure legends




**Supplemental Figure 1 related to Figure 1**

### Cholesterol is absent from testis lipid droplets.

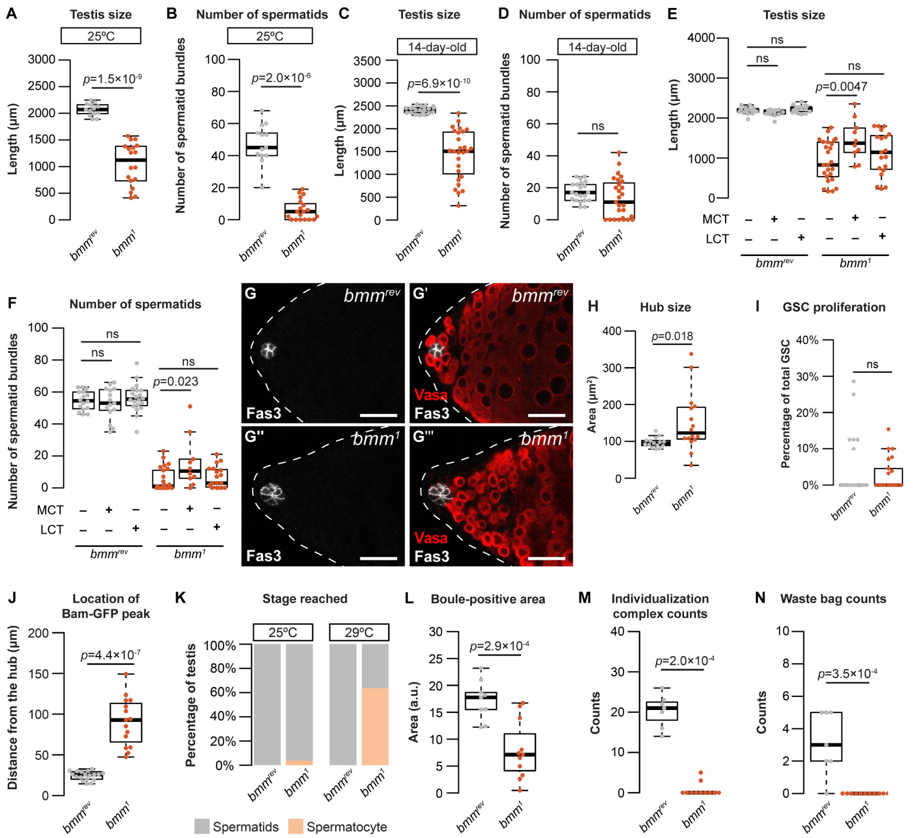
(A) Testes stained with BODIPY (A) to detect neutral lipids and Filipin III (A') to detect free cholesterol. Scale bars=50 μm.




### Supplemental Figure 2 related to Figure 2

## *bmm* mRNA levels during spermatogenesis in germline and somatic lineages.

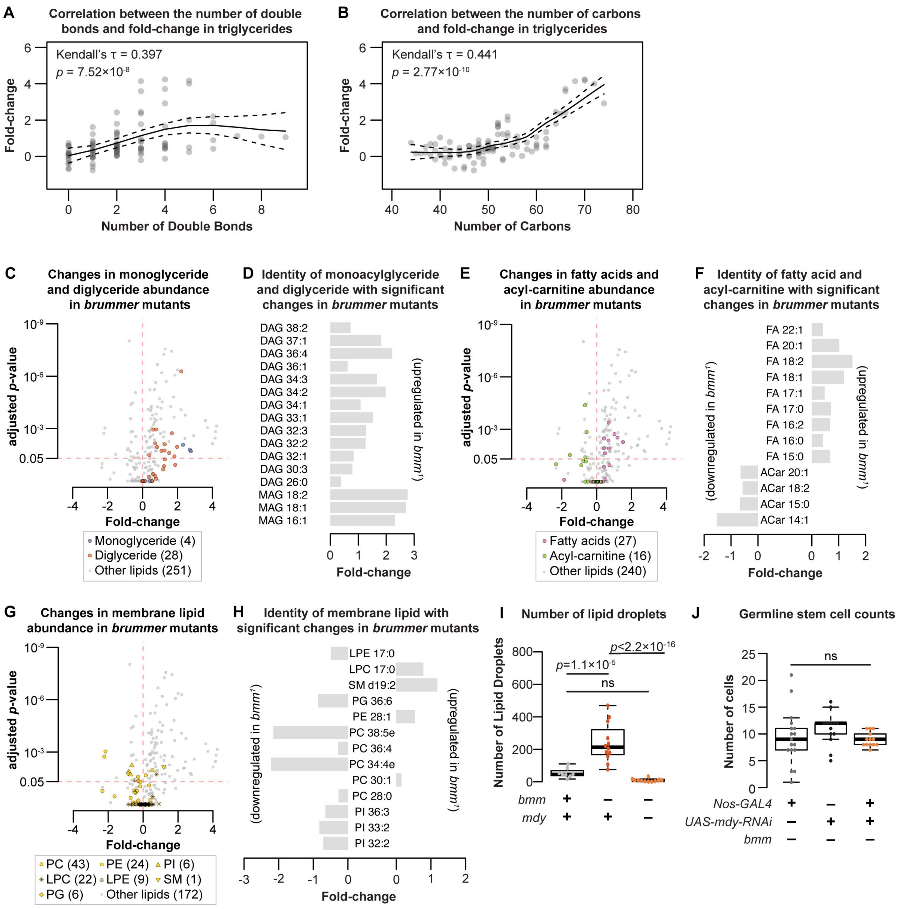
(A) Pseudotime trajectory of germline (black line) based on single-cell RNA sequencing data[62]. Individual cells are labeled according to the annotation within the data set. (B) Rolling average of normalized *bmm* transcript counts in the germline along the trajectory shown in panel A are plotted as a black line on the upper panel. Composition of cell types mapped on to the trajectory at each time point is shown at the lower half of panel B. (C) Pseudotime trajectory of the somatic cells (black line) based on publicly available single-cell RNA sequencing data[62]. Individual cells are labeled according to the annotation within the data set. (D) Rolling average of normalized *bmm* transcript counts in somatic cells plotted as a black line along the trajectory shown in C (upper panel). Composition of cell types mapped on to the trajectory at each time point (lower panel). (E-H) Representative images of $bmm^{rev}$ (E and F) and $bmm^{1}$ (G and H) testes with somatic over-expression of *GFP-LD* (*Tj-GAL4>UAS-GFP-LD*). Panel F and H contain magnified images of the area indicated by the boxes in panel E and G, respectively. In $bmm^{rev}$ testes, LD were restricted to a region near the apical tip (E) of the testis in both somatic (F–F''' arrows) and germline cells (F–F''' arrowheads). In $bmm^{1}$ testes, LD were present in both somatic (G–H arrows) and germline cells (G–H arrowheads), near the apical tip of the testis in a region corresponding to early-stage germ cells and in the region corresponding to spermatocytes. (E,G) Scale bars=50 μm; (F,H) scale bars=20 μm.




## Supplemental Figure 3 related to Figure 3

# Additional characterization of testis development and spermatogenesis in animals lacking *bmm*.

(A) Testis size was smaller in *bmm*$^{1}$ mutant animals compared with *bmm*$^{rev}$ controls at <24 hr post-eclosion when raised at 25°C (A; Welch two-sample t-test). (B) The number of spermatid bundles was significantly lower in *bmm*$^{1}$ mutant animals compared with *bmm*$^{rev}$ controls at <24 hr post-eclosion when raised at 25°C (Kruskal-Wallis rank sum test). (C) Testis size was significantly smaller in *bmm*$^{1}$ mutant males compared with *bmm*$^{rev}$ control males at 14-days post-eclosion (Welch two-sample t-test). (D) While the median number of spermatid bundles was not significantly different between *bmm*$^{1}$ mutant males and *bmm*$^{rev}$ control males at 14 days post-eclosion (Welch two-sample t-test), 8/27 *bmm*$^{1}$ testis had no spermatid bundles, a phenotype absent in age-matched *bmm*$^{rev}$ males (0/22) (p=0.0163, Pearson's Chi-squared test), suggesting a subtle defect is present. (E) Food supplemented with 4% medium chain triglyceride (MCT), but not long chain triglyceride (LCT), significantly increased testis length in *bmm*$^{1}$ animals but had no effect on this phenotype in *bmm*$^{rev}$ control animals (one-way ANOVA with Tukey multiple comparison test). (F) Food supplemented with 4% medium chain triglyceride significantly increased the number of spermatid bundles in *bmm*$^{1}$ testes but had no effect on this phenotype in *bmm*$^{rev}$ control animals (one-way ANOVA with Tukey multiple comparison test). (G) Representative images of *bmm*$^{rev}$ (G–G′) or *bmm*$^{1}$ (G′′– G′′′) testes stained for Fas3 (G and G′′) and Vas (G′ and G′′′). Scale bars=25 µm. (H) Quantification of hub area in *bmm*$^{rev}$ or *bmm*$^{1}$ testes showed a significantly larger hub size in *bmm*$^{1}$ testes (Welch two-sample t-test). (I) The number of germline stem cells (GSC) undergoing mitosis (phospho-histone H3$^{+}$ GSC/total GSC) was not significantly different between *bmm*$^{1}$ and *bmm*$^{rev}$ testes (Kruskal-Wallis rank sum test). (J) The distance between the hub and the first Bam-GFP positive cyst (Figure 3H) was significantly higher in *bmm*$^{1}$ testes than in *bmm*$^{rev}$ testes (Welch two-sample t-test). (K) All *bmm*$^{rev}$ testes and most *bmm*$^{1}$ testes contained spermatids when raised at 25°C; however, the most advanced stage of spermatogenesis observed in the majority of *bmm*$^{1}$ testes isolated from animals reared at 29°C was the spermatocyte stage. (L) Testes isolated from *bmm*$^{1}$ animals showed a significantly smaller Boule-positive area than control testes (Welch two-sample t-test). (M) Testes isolated from *bmm*$^{1}$ animals contain fewer individualization complexes than *bmm*$^{rev}$ control testes (Kruskal-Wallis rank sum test). (N) Fewer waste bags were present in testes isolated from *bmm*$^{1}$ animals compared with *bmm*$^{rev}$ control testes (Kruskal-Wallis rank sum test).




## Supplemental Figure 4 related to Figure 4

### Lipidomic analysis of animals lacking *bmm*.

(A) Higher fold-changes of triglycerides in *bmm*$^1$ animals were associated with less saturation in the acyl-groups (Kendall's rank correlation test). (B) Higher fold-changes of triglycerides in *bmm*$^1$ animals were associated with higher number of carbons in the acyl-groups (Kendall's rank correlation test). Each dot represents a single triglyceride species for panel B and C. (C) Volcano plot of identified lipids; monoglycerides shown in blue and diglycerides shown in orange. Many monoglycerides and diglycerides show increase in fold-change in *bmm*$^1$ males. (D) The number of carbon and the degree of saturation of monoglycerides (MAG) and diglycerides (DAG) with significant changes in abundance between *bmm*$^1$ and *bmm*$^{rev}$ males. (E) Volcano plot of identified lipids; fatty acids shown in magenta and acyl-carnitine shown in green. Many fatty acids show an increase in fold-change while many acyl-carnitines show a decrease in fold-change in *bmm*$^1$ males. (F) The number of carbon and the degree of saturation of fatty acids (FA) and acyl-carnitines (ACar) with significant changes in abundance between *bmm*$^1$ and *bmm*$^{rev}$ males. (G) Volcano plot of identified lipids; membrane lipids shown in yellow. (H) The number of carbon and the degree of saturation of membrane lipids with significant changes in abundance between *bmm*$^1$ and *bmm*$^{rev}$ males. For panel G and H, PC: phosphatidylcholine; PE: phosphatidylethanolamine; PI: phosphatidylinositol; LPC: lysophosphatidylcholine; LPE: lysophosphatidylethanolamine; SM: sphingomyelin; PG: phosphatidylglycerol. (I) Loss of *mdy* function rescued the elevated number of LD in *bmm*$^1$ testes to control levels (one-way ANOVA with Tukey multiple comparison test). (J) Germline-specific loss of *mdy* in *bmm*$^1$ animals did not reduce GSC numbers, but the variance in GSC number was significantly rescued (*nos-GAL4*>+; *bmm*$^1$ vs *nos-GAL4>mdy RNAi*; *bmm*$^1$: $p$=4.5 × 10$^{-5}$; +>*mdy RNAi*; *bmm*$^1$ vs *nos-GAL4>mdy RNAi*; *bmm*$^1$: $p$=0.0082 by F-test).

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## Reviewer #1 (Public Review):

In this study, the authors investigate the role of triglycerides in spermatogenesis. This work is based on their previous study (PMID: 31961851) on triglyceride sex differences in which they showed that somatic testicular cells play a role in whole body triglyceride homeostasis. In the current study, they show that lipid droplets (LDs) are significantly higher in the stem and progenitor cell (pre-meiotic) zone of the adult testis than in the meiotic spermatocyte stages. The distribution of LDs anti-correlates with the expression of the triglyceride lipase Brummer (Bmm), which has higher expression in spermatocytes than early germline stages. Analysis of a bmm mutant (bmm[1]) - a P-element insertion that is likely a hypomorphic - and its revertant (bmm[rev]) as a control shows that bmm acts autonomously in the germline to regulate LDs. In particular, the number of LDs is significantly higher in spermatocytes from bmm[1] mutants than from bmm[rev] controls. Testes from males with global loss of bmm (bmm[1]) are shorter than controls and have fewer differentiated spermatids. The zone of bam expression, typically close to the niche/hub in WT, is now many cell diameters away from the hub in bmm[1] mutants. There is an increase in the number of GSCs in bmm[1] homozygotes, but this phenotype is probably due to the enlarged hub. However, clonal analyses of GSCs lacking bmm indicate that a greater percentage of the GSC pool is composed of bmm[1]-mutant clones than of bmm[rev]-clones. This suggests that loss of bmm could impart a competitive advantage to GSCs, but this is not explored in greater detail. Despite the increase in number of GSCs that are bmm[1]-mutant clones, there is a significant reduction in the number of bmm[1]-mutant spermatocyte and post-meiotic clones. This suggests that fewer bmm[1]-mutant germ cells differentiate than controls. To gain insights into triglyceride homeostasis in the absence of bmm, they perform mass spec-based lipidomic profiling. Analyses of these data support their model that triglycerides are the class of lipid most affected by loss of bmm, supporting their model that excess triglycerides are the cause of spermatogenetic defects in bmm[1]. Consistent with their model, a double mutant of bmm[1] and a diacylglycerol O-acyltransferase 1 called midway (mdy) reverts the bmm-mutant germline phenotypes.

There are numerous strengths of this paper. First, the authors report rigorous measurements and statistical analyses throughout the study. Second, the authors utilize robust genetic analyses with loss-of-function mutants and lineage-specific knockdown. Third, they demonstrate the appropriate use of controls and markers. Fourth, they show rigorous lipidomic profiling. Lastly, their conclusions are appropriate for the results. In other words, they don't overstate the results.

There are a few weaknesses. Although the results support the germline autonomous role of bmm in spermatogenesis, one potential caveat that the mdy rescue was global, i.e., in both somatic and germline lineages. The authors did not recover somatic bmm clones, suggesting that bmm may be required for somatic stem self-renewal and/or niche residency. While this is beyond the scope of this paper, it is possible that somatic bmm does impact germline differentiation in a global bmm mutant. Regarding data presentation, I have a minor point about Fig. 3L: why aren't all data shown as box plots (only Day 14 bmm[rev] does). Finally, the authors provide a detailed pseudotime analysis of snRNA-seq of the testis in Fig. S2A-D, but this analysis is not sufficiently discussed in the text.

Overall, the many strengths of this paper outweigh the relatively minor weaknesses. The rigorously quantified results support the major aim that appropriate regulation of triglycerides are needed in a germline cell-autonomous manner for spermatogenesis.

This paper should have a positive impact on the field. First and foremost, there is limited knowledge about the role of lipid metabolism in spermatogenesis. The lipidomic data will be useful to researchers in the field who study various lipid species. Going forward, it will be very interesting to determine what triglycerides regulate in germline biology. In other words, what functions/pathways/processes in germ cells are negatively impacted by elevated triglycerides. And as the authors point out in the discussion, it will be important to determine what regulates bmm expression such that bmm is higher in later stages of germline differentiation.

## Reviewer #2 (Public Review):

Summary:

Here, the authors show that neutral lipids play a role in spermatogenesis. Neutral lipids are components of lipid droplets, which are known to maintain lipid homeostasis, and to be involved in non-gonadal differentiation, survival, and energy. Lipid droplets are present in the testis in mice and Drosophila, but not much is known about the role of lipid droplets during spermatogenesis. The authors show that lipid droplets are present in early differentiating germ cells, and absent in spermatocytes. They further show a cell autonomous role for the lipase brummer in regulating lipid droplets and, in turn, spermatogenesis in the Drosophila testis. The data presented show that a relationship between lipid metabolism and spermatogenesis is congruous in mammals and flies, supporting Drosophila spermatogenesis as an effective model to uncover the role lipid droplets play in the testis.

Strengths and weaknesses:

The authors do a commendably thorough characterization of where lipid droplets are detected in normal testes: located in young somatic cells, and early differentiating germ cells. They use multiple control backgrounds in their analysis, including w[1118], Canton S, and Oregon R, which adds rigor to their interpretations. The authors employ markers that identify which lipid droplets are in somatic cells, and which are in germ cells. The authors use these markers to present measured distances of somatic and germ cell-derived lipid droplets from the hub. Because they can also measure the distance of somatic and germ cells with age-specific markers from the hub, these results allow the authors to correlate position of lipid droplets with the age of cells in which they are present. This analysis is clearly shown and well quantified.

The quantification of lipid droplet distance from the hub is applied well in comparing brummer mutant testes to wild type controls. The authors measure the number of lipid droplets of specific diameters, and the spatial distribution of lipid droplets as a function of distance from the hub. These measurements quantitatively support their findings that lipid droplets are present in an expanded population of cells further from the hub in brummer mutants. The authors further quantify lipid droplets in germline clones of specified ages; the quantitative analysis here is displayed clearly, and supports a cell autonomous role for brummer in regulating lipid droplets in spermatocytes.

Data examining testis size and number of spermatids in brummer mutants clearly indicates the importance of regulating lipid droplets to spermatogenesis. The authors show beautiful images supported by rigorous quantification supporting their findings that brummer mutants have both smaller testes with fewer spermatids at both 29 and 25C. There is also significant data supporting defects in testis size for 14-day-old brummer mutant animals compared to controls. The comparison of number of spermatids at this age is not significant, which does not detract from the the story but does not support sperm development defects specifically caused by brummer loss at 14 days. Their analysis clearly shows an expanded region beyond the testis apex that includes younger germ cells, supporting a role for lipid droplets influencing germ cell differentiation during spermatogenesis.

The authors present a series of data exploring a cell autonomous role for brummer in the germline, including clonal analysis and tissue specific manipulations. The clonal data indicating increased lipid droplets in spermatocyte clones, and a higher proportion of brummer mutant GSCs at the hub are convincing and supported by quantitation. The authors also show a tissue specific rescue of the brummer testis size phenotype by knocking down mdy specifically in germ cells, which is also supported by statistically significant quantitation. The authors present data examining the number of spermatocyte and post-meiotic clones 14 days after clonal induction. While data they present is significant with a 95% confidence interval and a p value of 0.0496, its significance is not as robust as other values reported in the study, and it is unclear how much information can be gained from that specific result.

The authors do a beautiful job of validating where they detect brummer-GFP by presenting their own pseudotime analysis of publicly available single cell RNA sequencing data. Their data is presented very clearly, and supports expression of brummer in older somatic and germline cells of the age when lipid droplets are normally not detected. The authors also present a thorough lipidomic analysis of animals lacking brummer to identify triglycerides as an important lipid droplet component regulating spermatogenesis.

Impact:

The authors present data supporting the broad significance of their findings across phyla. This data represents a key strength of this manuscript. The authors show that loss of a conserved triglyceride lipase impacts testis development and spermatogenesis, and that these impacts can be rescued by supplementing diet with medium-chain triglycerides. The authors point out that these findings represent a biological similarity between Drosophila and mice, supporting the relevance of the Drosophila testis as a model for understanding the role of lipid droplets in spermatogenesis. The connection buttresses the relevance of these findings and this model to a broad scientific community.

## Reviewer #3 (Public Review):

In this manuscript, Chao et al seek to understand the role of brummer, a triglyceride lipase, in the Drosophila testis. They show that Brummer regulates lipid droplet degradation during differentiation of germ and somatic cells, and that this process is essential for normal development to progress. These findings are interesting and novel, and contribute to a growing realisation that lipid biology is important for differentiation.

Major comments:

1. The data in Figs 1 and 2, while helpful in setting the scene, do not add much to what was previously shown by the same group, namely that lipid droplets are present in both early germ cells and early somatic cells in the testis, and that Bmm regulates their degradation (PMID: 31961851). Measuring the distance of lipid droplets from the hub, while helpful in quantifying what is apparent, that only stem and early differentiated stages have lipid droplets, is not as informative as the way data are presented later (Fig. 2I), where droplets in specific stages are measured. Much of this could be condensed without much overall loss to the manuscript.

2. It would be important to show images of the clones from which the data in Fig. 2I are generated. The main argument is that Bmm regulates lipid droplets in a cell autonomous manner; these data are the strongest argument in support of this and

should be emphasised at the expense of full animal mutants (which could be moved to supplementary data). Similarly, the title of Fig. S2 ("brummer regulates lipid droplets in a cell autonomous manner") should be changed as the figure has no experiments with cell (or cell-type)-specific knockdowns/mutants. This figure does show changes in lipid droplets in both lineages in bmm mutants, so an appropriate title could be "brummer regulates lipid droplets in both germ and soma".

3. Interestingly, the clonal data show that bmm is dispensable in germ cells until spermatocyte stages, as no increase in lipid droplet number is seen until then. This should be more clearly stated, as it indicates that the important function of Bmm is to degrade lipid droplets at the transition from spermatogonial to spermatocyte stages. This is consistent with the phenotypes observed in which late stage germ cells are reduced or missing. However, the effect on niche retention of the mutant GSCs at the expense of neighbouring wildtype GSCs is hard to explain. Are lipid droplets in mutant GSCs larger than in control? Is there any discernible effect of bmm mutation on lipids in GSCs? Additionally, bam expression is delayed, suggesting that bmm may have roles on cell fate in earlier stages than its roles that can be detected on lipid droplets.

4. The bmm loss-of-function phenotype could be better described. Some of the data is glossed over with little description in the text (see for example the reference to Fig. 3A-C). For instance, in the discussion, the text states "loss of bmm delays germline differentiation leading to an accumulation of early-stage germ cells" (p13, l.259-60). However, this accumulation has not been clearly shown, or at least described in the manuscript. Most of the data show a reduction (or almost complete absence) of differentiated cell types. This could indeed be due to delayed differentiation, or alternatively to a block in differentiation or to death of the differentiated cells. The clonal data presented show a decrease in the number of cells recovered, but do not allow inferences as to the timing of differentiation, making it hard to distinguish between the various possibilities for the lack of differentiated spermatids. Apart from data showing that GSCs are more likely to remain at the niche, no further data are shown to support the fact that mutant germ cells accumulate in early stages. While additional experiments could help resolve some of these issues, much of this could also be resolved by tempering the conclusions drawn in the text.

5. In the discussion (p.14, l-273 onwards), the authors suggest that products of triglyceride breakdown are important for spermatogenesis. However, an alternative interpretation of the results presented here (especially those using the midway mutant) could be that triglycerides impede normal differentiation directly. Indeed, preventing the cells' ability to produce triglycerides in the first place can rescue many of the defects observed. A better discussion of these results with a model for the function of triglycerides and their by-products would be a great improvement to this manuscript.

## Author Response

We would like to extend our thanks to the reviewers who took the time to carefully read our paper and provide thoughtful insights and suggestions on how to strengthen our conclusions. All reviewers agreed that our study presented strong data supporting a role for triglyceride lipase brummer (bmm) in regulating testis lipid droplets and spermatogenesis in Drosophila, and that our findings advance our understanding of lipid biology during sperm development. Reviewers also made several helpful suggestions on how to strengthen our manuscript even further. Below, we provide a brief outline of our plans to revise this manuscript in response to reviewer comments.

The majority of reviewer comments will be addressed by text changes, rearranging figures to add images, and making a model to visually represent our findings. Together, these changes will ensure we clearly communicate our data and conclusions with readers, and properly contextualize our findings. See below for details on our planned revisions.

> ### *Reviewer #1 (Public Review):*

> *In this study, the authors investigate the role of triglycerides in spermatogenesis. This work is based on their previous study (PMID: 31961851) on triglyceride sex differences in which they showed that somatic testicular cells play a role in whole body triglyceride homeostasis. In the current study, they show that lipid droplets (LDs) are significantly higher in the stem and progenitor cell (pre-meiotic) zone of the adult testis than in the meiotic spermatocyte stages. The distribution of LDs anti-correlates with the expression of the triglyceride lipase Brummer (Bmm), which has higher expression in spermatocytes than early germline stages. Analysis of a bmm mutant (bmm[1]) - a P-element insertion that is likely a hypomorphic - and its revertant (bmm[rev]) as a control shows that bmm acts autonomously in the germline to regulate LDs. In particular, the number of LDs is significantly higher in spermatocytes from bmm[1] mutants than from bmm[rev] controls. Testes from males with global loss of bmm (bmm[1]) are shorter than controls and have fewer differentiated spermatids. The zone of bam expression, typically close to the niche/hub in WT, is now many cell diameters away from the hub in bmm[1] mutants. There is an increase in the number of GSCs in bmm[1] homozygotes, but this phenotype is probably due to the enlarged hub. However, clonal analyses of GSCs lacking bmm indicate that a greater percentage of the GSC pool is composed of bmm[1]-mutant clones than of bmm[rev]-clones. This suggests that loss of bmm could impart a competitive advantage to GSCs, but this is not explored in greater detail. Despite the increase in number of GSCs that are bmm[1]-mutant clones, there is a significant reduction in the number of bmm[1]-mutant spermatocyte and post-meiotic clones. This suggests that fewer bmm[1]-mutant germ cells differentiate than controls. To gain insights into triglyceride homeostasis in the absence of bmm, they perform mass spec-based lipidomic profiling. Analyses of these data support their model that triglycerides are the class of lipid most affected by loss of bmm, supporting their model that excess triglycerides are the cause of spermatogenetic defects in bmm[1]. Consistent with their model, a double mutant of bmm[1] and a diacylglycerol O-acyltransferase 1 called midway (mdy) reverts the bmm-mutant germline phenotypes.*

> *There are numerous strengths of this paper. First, the authors report rigorous measurements and statistical analyses throughout the study. Second, the authors utilize robust genetic analyses with loss-of-function mutants and lineage-specific knockdown. Third, they demonstrate the appropriate use of controls and markers. Fourth, they show rigorous lipidomic profiling. Lastly, their conclusions are appropriate for the results. In other words, they don't overstate the results.*

We thank the Reviewer for their positive assessment of our paper.

> *There are a few weaknesses. Although the results support the germline autonomous role of bmm in spermatogenesis, one potential caveat that the mdy rescue was global, i.e., in both somatic and germline lineages. The authors did not recover somatic bmm clones, suggesting that bmm may be required for somatic stem self-renewal and/or niche residency. While this is beyond the scope of this paper, it is possible that somatic bmm does impact germline differentiation in a global bmm mutant.*

In the revised manuscript, we will more clearly delineate when we used global versus germline-only loss of mdy to rescue bmm mutant phenotypes in the testis. We will also acknowledge the possibility that somatic bmm may play a role in germline differentiation in a global bmm mutant.

> *Regarding data presentation, I have a minor point about Fig. 3L: why aren't all data shown as box plots (only Day 14 bmm[rev] does). Finally, the authors provide a detailed pseudotime analysis of snRNA-seq of the testis in Fig. S2A-D, but this analysis is not sufficiently discussed in the text.*

We will make text and presentation changes in the revised manuscript to describe our data more clearly, and will add text to describe our pseudotime analysis of single-cell RNA seq data in more detail.

> *Overall, the many strengths of this paper outweigh the relatively minor weaknesses. The rigorously quantified results support the major aim that appropriate regulation of triglycerides are needed in a germline cell-autonomous manner for spermatogenesis.*

> *This paper should have a positive impact on the field. First and foremost, there is limited knowledge about the role of lipid metabolism in spermatogenesis. The lipidomic data will be useful to researchers in the field who study various lipid species. Going forward, it will be very interesting to determine what triglycerides regulate in germline biology. In other words, what functions/pathways/processes in germ cells are negatively impacted by elevated triglycerides. And as the authors point out in the discussion, it will be important to determine what regulates bmm expression such that bmm is higher in later stages of germline differentiation.*

We agree with the reviewer about the many interesting future directions for this project. We will therefore add a model figure in the revised manuscript to visualize our findings and highlight remaining questions about how bmm and triglycerides support normal spermatogenesis in Drosophila.

> ## *Reviewer #2 (Public Review):*

> *Summary:*

> *Here, the authors show that neutral lipids play a role in spermatogenesis. Neutral lipids are components of lipid droplets, which are known to maintain lipid homeostasis, and to be involved in non-gonadal differentiation, survival, and energy. Lipid droplets are present in the testis in mice and Drosophila, but not much is known about the role of lipid droplets during spermatogenesis. The authors show that lipid droplets are present in early differentiating germ cells, and absent in spermatocytes. They further show a cell autonomous role for the lipase brummer in regulating lipid droplets and, in turn, spermatogenesis in the Drosophila testis. The data presented show that a relationship between lipid metabolism and spermatogenesis is congruous in mammals and flies, supporting Drosophila spermatogenesis as an effective model to uncover the role lipid droplets play in the testis.*

We thank the Reviewer for their positive assessment of our paper.

> *Strengths and weaknesses:*

> *The authors do a commendably thorough characterization of where lipid droplets are detected in normal testes: located in young somatic cells, and early differentiating germ cells. They use multiple control backgrounds in their analysis, including w[1118], Canton S, and Oregon R, which adds rigor to their interpretations. The authors employ markers that identify which lipid droplets are in somatic cells, and which are in germ cells. The authors use these markers to present measured distances of somatic and germ cell-derived lipid droplets from the hub. Because they can also measure the distance of somatic and germ cells with age-specific markers from the hub, these results allow the authors to correlate position of lipid droplets with the age of cells in which they are present. This analysis is clearly shown and well quantified.*

> *The quantification of lipid droplet distance from the hub is applied well in comparing brummer mutant testes to wild type controls. The authors measure the number of lipid droplets of specific diameters, and the spatial distribution of lipid droplets as a function of distance from the hub. These measurements quantitatively support their findings that lipid droplets are present in an expanded population of cells further from the hub in brummer mutants. The authors further quantify lipid droplets in germline clones of specified ages; the quantitative analysis here is displayed clearly, and supports a cell autonomous role for brummer in regulating lipid droplets in spermatocytes.*

> *Data examining testis size and number of spermatids in brummer mutants clearly indicates the importance of regulating lipid droplets to spermatogenesis. The authors show beautiful images supported by rigorous quantification supporting their findings that brummer mutants have both smaller testes with fewer spermatids at both 29 and 25C. There is also significant data supporting defects in testis size for 14-day-old brummer mutant animals compared to controls. The comparison of number of spermatids at this age is not significant, which does not detract from the the story but does not support sperm development defects specifically caused by brummer loss at 14 days. Their analysis clearly shows an expanded region beyond the testis apex that includes younger germ cells, supporting a role for lipid droplets influencing germ cell differentiation during spermatogenesis.*

We thank the reviewer for pointing out this inaccuracy in our manuscript. In the revised manuscript we will choose more precise language to describe defects in sperm development in 14-day-old bmm mutants.

> *The authors present a series of data exploring a cell autonomous role for brummer in the germline, including clonal analysis and tissue specific manipulations. The clonal data indicating increased lipid droplets in spermatocyte clones, and a higher proportion of brummer mutant GSCs at the hub are convincing and supported by quantitation. The authors also show a tissue specific rescue of the brummer testis size phenotype by knocking down mdy specifically in germ cells, which is also supported by statistically significant quantitation. The authors present data examining the number of spermatocyte and post-meiotic clones 14 days after clonal induction. While data they present is significant with a 95% confidence interval and a p value of 0.0496, its significance is not as robust as other values reported in the study, and it is unclear how much information can be gained from that specific result.*

We thank the reviewer for raising this point. In the revised manuscript we will display the p-value clearly to ensure our statistical output is clear for readers to evaluate our conclusions regarding bmm mutant clones 14 days after clone induction.

> *The authors do a beautiful job of validating where they detect brummer-GFP by presenting their own pseudotime analysis of publicly available single cell RNA sequencing data. Their data is presented very clearly, and supports expression of brummer in older somatic and germline cells of the age when lipid droplets are normally not detected. The authors also present a thorough lipidomic analysis of animals lacking brummer to identify triglycerides as an important lipid droplet component regulating spermatogenesis.*

> *Impact:*

> *The authors present data supporting the broad significance of their findings across phyla. This data represents a key strength of this manuscript. The authors show that loss of a conserved triglyceride lipase impacts testis development and spermatogenesis, and that these impacts can be rescued by supplementing diet with medium-chain triglycerides. The authors point out that these findings represent a biological similarity between Drosophila and mice, supporting the relevance of the Drosophila testis as a model for understanding the role of lipid droplets in spermatogenesis. The connection buttresses the relevance of these findings and this model to a broad scientific community.*

> ### *Reviewer #3 (Public Review):*

> *In this manuscript, Chao et al seek to understand the role of brummer, a triglyceride lipase, in the Drosophila testis. They show that Brummer regulates lipid droplet degradation during differentiation of germ and somatic cells, and that this process is essential for normal development to progress. These findings are interesting and novel, and contribute to a growing realisation that lipid biology is important for differentiation.*

We thank the Reviewer for their positive assessment of our manuscript.

> *Major comments:*

> 1. *The data in Figs 1 and 2, while helpful in setting the scene, do not add much to what was previously shown by the same group, namely that lipid droplets are present in both early germ cells and early somatic cells in the testis, and that Bmm regulates their degradation (PMID: 31961851). Measuring the distance of lipid droplets from the hub, while helpful in quantifying what is apparent, that only stem and early differentiated stages have lipid droplets, is not as informative as the way data are presented later (Fig. 2I), where droplets in specific stages are measured. Much of this could be condensed without much overall loss to the manuscript.*

We thank the reviewer for this comment and will condense the first part of the paper in our revised manuscript.

> 2. *It would be important to show images of the clones from which the data in Fig. 2I are generated. The main argument is that Bmm regulates lipid droplets in a cell autonomous manner; these data are the strongest argument in support of this and should be emphasised at the expense of full animal mutants (which could be moved to supplementary data).*

We thank the reviewer for this comment, and will add an image in our revised manuscript showing lipid droplets in bmm mutant spermatocyte clones.

> *Similarly, the title of Fig. S2 ("brummer regulates lipid droplets in a cell autonomous manner") should be changed as the figure has no experiments with cell (or cell-type)-specific knockdowns/mutants. This figure does show changes in lipid droplets in both lineages in bmm mutants, so an appropriate title could be "brummer regulates lipid droplets in both germ and soma".*

We thank the reviewer for this comment, and will adjust the S2 figure legend title in the revised manuscript.

> 3. *Interestingly, the clonal data show that bmm is dispensable in germ cells until spermatocyte stages, as no increase in lipid droplet number is seen until then. This should be more clearly stated, as it indicates that the important function of Bmm is to degrade lipid droplets at the transition from spermatogonial to spermatocyte stages. This is consistent with the phenotypes observed in which late stage germ cells are reduced or missing. However, the effect on niche retention of the mutant GSCs at the expense of neighbouring wildtype GSCs is hard to explain. Are lipid droplets in mutant GSCs larger than in control? Is there any discernible effect of bmm mutation on lipids in GSCs? Additionally, bam expression is delayed, suggesting that bmm may have roles on cell fate in earlier stages than its roles that can be detected on lipid droplets.*

We thank the reviewer for this comment. We will include more text in the revised manuscript to clarify the key role bmm plays in regulating lipid droplets at the spermatogonia-spermatocyte transition. We will also add more detail and potentially data to our description of how bmm affects lipid droplets in cells at the earliest stages of germline development.

> 4. *The bmm loss-of-function phenotype could be better described. Some of the data is glossed over with little description in the text (see for example the reference to Fig. 3A-C). For instance, in the discussion, the text states "loss of bmm delays germline differentiation leading to an accumulation of early-stage germ cells" (p13, l.259-60). However, this accumulation has not been clearly shown, or at least described in the manuscript. Most of the data show a reduction (or almost complete absence) of differentiated cell types. This could indeed be due to delayed differentiation, or alternatively to a block in differentiation or to death of the differentiated cells. The clonal data presented show a decrease in the number of cells recovered, but do not allow inferences as to the timing of differentiation, making it hard to distinguish between the various possibilities for the lack of differentiated spermatids. Apart from data showing that GSCs are more likely to remain at the niche, no further data are shown to support the fact that mutant germ cells accumulate in early stages. While additional experiments could help resolve some of these issues, much of this could also be resolved by tempering the conclusions drawn in the text.*

We thank the reviewer for these comments. In the revised manuscript we will temper our conclusions regarding bmm’s precise role in spermatogenesis by discussing different mechanisms (e.g. differentiation or death) that could lead to the phenotypes we observe.

> 5. *In the discussion (p.14, l-273 onwards), the authors suggest that products of triglyceride breakdown are important for spermatogenesis. However, an alternative interpretation of the results presented here (especially those using the midway mutant) could be that triglycerides impede normal differentiation directly. Indeed, preventing the cells' ability to produce triglycerides in the first place can rescue many of the defects observed. A better discussion of these results with a model for the function of triglycerides and their by-products would be a great improvement to this manuscript.*

We thank the reviewer for this comment. To ensure our data is clearly communicated with readers, we will add a model to the paper suggesting how triglyceride and its by-products influence spermatogenesis.

Together, these changes will strengthen our overall finding that bmm-mediated regulation of testis triglyceride is important for normal sperm development. Because our findings in flies align with and extend data from rodent models, the developmental mechanisms we uncovered about how triglyceride lipase bmm regulates testis lipid droplets and sperm development will likely operate in other species.